

CHEMICAL REACTIONS

Endothermic vs Exothermic

Detailed Lesson Plan Grade 4 | Fun Science Series

Subject	Chemistry (Grade 4 Science)
Duration	45-60 minutes
Learning Goals	Classify reactions; identify heat flow direction
Materials Needed	Baking soda, vinegar, thermometer, clear plastic cups
Assessment	Interactive Sorting Activity + Exit Ticket individual quiz

SECTION 1: LEARNING OBJECTIVES

1. Define what a chemical reaction is and give everyday examples.
2. Explain the clear differences between exothermic and endothermic reactions.
3. Use practical memory hooks (**Exo = Exit** and **Endo = Enter**) to classify reactions quickly.
4. Identify visual or sensory signal words (gets hot, absorbs heat, feels cold) within descriptions.
5. Correctly sort at least 4 out of 6 example reactions into the right category during the lesson.
6. Make meaningful real-world connections to daily Indian life and familiar cultural contexts.

SECTION 2: BACKGROUND KNOWLEDGE FOR TEACHERS

What Is a Chemical Reaction?

A chemical reaction is a process in which one or more substances (called **reactants**) are converted into one or more new substances (called **products**). The key indicator is that a **new substance is formed** with entirely different properties from the starting materials. This process is written and represented as:

Reactants → Products

Common misconception to address: Physical changes like melting ice or dissolving sugar are **NOT** chemical reactions because no new substance is formed. In physical changes, the original substance remains chemically identical and can be completely recovered.

Energy and Chemical Reactions

All chemical reactions involve an exchange of energy with their surroundings, usually in the form of heat. Whether the surroundings feel warmer or cooler depends completely on the direction of that energy flow:

Exothermic Reactions	Endothermic Reactions
Heat is released to the surroundings.	Heat is absorbed from the surroundings.
Surroundings feel HOT .	Surroundings feel COLD .
Energy of products < Energy of reactants.	Energy of products > Energy of reactants.
Examples: combustion, rusting, respiration reactions.	Examples: photosynthesis, evaporation processes.

SECTION 3: DETAILED LESSON PLAN

Phase 1: Warm-Up / Hook (5-7 min)

- Begin with a relatable, sensory question: *"Why does your hand feel warm when you rub it fast together? Why does an ice or cold pack feel instantly cold on your skin?"*
- Ask students to think of exactly one thing that gets **HOT** and one thing that gets **COLD** in their daily lives.
- Record student responses dynamically on the board under two designated columns: **HOT** and **COLD**.
- Tell students: *"Today we're going to solve the scientific mystery of why some reactions make things hot and others make things cold!"*

Phase 2: Direct Instruction: Chemical Reactions (8-10 min)

- **Define:** A chemical reaction happens when starting substances completely change into something entirely new.
- Write clearly on the blackboard: **Reactants** → **Products**.
- Give rich Indian context examples: Roti cooking on a tawa, a burning diya, a rusting old iron gate, curd forming from warm milk, and baking a cake.
- Emphasize the foundational rule: A new substance must form. Melting ice = NOT a reaction (it's still water). Dissolving sugar = NOT a reaction.
- **Quick check:** Ask 2-3 students to give a unique example and say out loud if it represents a chemical reaction.

Phase 3: Exothermic Reactions (8 min)

- Introduce the term: **Exothermic**. Write on the board: *Exo = Exit → Heat Exits → Gets HOT*.
- Signal words to recognize exothermic reactions: *gets hot, burning, flame, releases heat, gives off heat*.
- Real-life examples from the lesson: Burning coal in a sigri (keeps us warm in winter), a rusting iron gate (releases heat very slowly), and cooking dal on a gas stove.
- Ask students: "*Can you think of something at home that gets HOT when a change happens?*"
- **Optional demo:** Hold hands together and rub them vigorously to demonstrate that heat is produced (exothermic friction analogy).

Phase 4: Endothermic Reactions (8 min)

- Introduce the term: **Endothermic**. Write on the board: *Endo = Enter → Heat Enters → Gets COLD*.
- Signal words to recognize endothermic reactions: *gets cold, absorbs heat, cools down, needs heat input*.
- Real-life examples: A clay pot (matka) cooling water, baking soda + vinegar mixed together (glass feels cold), and photosynthesis (plants absorbing sunlight).
- Ask students: "*Why does water stay cool in a clay pot? What does the pot do with the heat?*"
- Connect to home: Ice packs used on injuries—they absorb heat away from the body to make it feel cold.

Phase 5: Sorting Activity (10 min)

- Present the 6 core sorting challenges from the lesson (listed sequentially in Section 5 below).
- Students can work productively in pairs. Give each pair a physical sorting card or have them write in their notebooks.
- Circulate around the room and guide students who are unsure, prompting them with: "*Does it naturally get hot or cold?*"
- Review all answers together as a class, clearly explaining the reasoning for each one.

Phase 6: Recap & Exit Ticket (7-10 min)

- Do a quick class-wide recap: Chemical reaction = new substance; *Exo = Exit = HOT; Endo = Enter = COLD*.
 - Hand out the exit ticket assessment (see Section 6)—3 quick questions to check for concrete understanding.
 - Assign the **Science Spy Challenge** homework: Find ONE hot or cold reaction at home and record it.
 - Preview the next upcoming lesson: "*Next time, we'll do REAL experiments—get ready!*"
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SECTION 4: KEY CONCEPTS & MEMORY HOOKS

EXOTHERMIC: *EXO* = *EXIT* • Heat **EXITS** the reaction • Surroundings get **HOT**

ENDOTHERMIC: *ENDO* = *ENTER* • Heat **ENTERS** the reaction • Surroundings get **COLD**

Comprehensive Comparison Table

Feature	Exothermic Reactions	Endothermic Reactions
Heat Flow	Released (OUT) to surroundings	Absorbed (IN) from surroundings
Temperature Effect	Surroundings get WARMER	Surroundings get COOLER
Prefix Meaning	<i>Exo</i> = Exit	<i>Endo</i> = Enter
Energy of Products	Lower than starting reactants	Higher than starting reactants
Indian Life Examples	Burning diya Coal burning in a sigri	Clay pot (matka) cooling water Plants doing photosynthesis
Kitchen Examples	Cooking dal on a gas stove	Baking soda + vinegar (feels cold)
Signal Words	Hot, burning, flame, releases, gives off	Cold, absorbs, cools down, needs heat

SECTION 5: CLASSROOM SORTING ACTIVITY

Instruct students to classify each reaction below as either **Exothermic** or **Endothermic**. Ask them to underline the signal word(s) that helped them decide.

#	Reaction Description	Answer	Reason / Signal Word
1	Burning Diwali crackers produces light and heat.	Exothermic	Signal: produces heat, burning
2	Cold pack placed on a sprained ankle feels cold.	Endothermic	Signal: feels cold, absorbs heat from body
3	Wood burning in Lohri bonfire gives off warmth.	Exothermic	Signal: gives off warmth, burning, flame

#	Reaction Description	Answer	Reason / Signal Word
4	Plants making food using sunlight (photosynthesis).	Endothermic	Signal: absorbs sunlight (energy input needed)
5	Cooking dal on the gas stove pot gets very hot.	Exothermic	Signal: gets hot, burning gas
6	Baking soda + vinegar mixed in a glass — glass feels cold.	Endothermic	Signal: feels cold, absorbs heat during reaction
7	Rusting of an old iron gate left outside in rain.	Exothermic	Releases heat slowly; a slow exothermic reaction
8	Curd (yoghurt) forming from warm milk.	Exothermic	Fermentation releases heat; milk gets slightly warm

**Bonus questions for advanced students.*

SECTION 6: EXIT TICKET ASSESSMENT

Distribute at the end of class. Students answer individually in 5 minutes. Use responses to identify students who need more support.

Name:

Date:

Class:

Q1. What is a chemical reaction? Write one sentence.

Answer:

Q2. Circle the correct answer:

- An exothermic reaction: **a)** Absorbs heat **b)** Releases heat **c)** Does not involve heat
- An endothermic reaction: **a)** Gets hot **b)** Gets cold **c)** Stays the same

Q3. Classify each reaction and write EXO or ENDO:

a) Burning a candle: _____

b) Ice pack on knee: _____

c) Cooking roti on tawa: _____

d) Mixing baking soda + vinegar: _____

Q4. Complete the memory hooks:

- Exo = _____ → Heat _____ → Gets _____
- Endo = _____ → Heat _____ → Gets _____

Q5. (Bonus) Give one example of an exothermic reaction from your daily life that was NOT mentioned in class.

Answer:

Answer Key (Teacher Copy)

- **Q1:** A chemical reaction is when substances change into a completely new substance. (*Accept any reasonable definition*).
 - **Q2:** Exothermic: **b)** Releases heat | Endothermic: **b)** Gets cold.
 - **Q3:** **a)** EXO, **b)** ENDO, **c)** EXO, **d)** ENDO.
 - **Q4:** Exo = Exit = Heat Exits = Gets HOT | Endo = Enter = Heat Enters = Gets COLD.
 - **Q5:** *Accept:* Burning wood, campfire, sparkler, digestion of food, hand warmers, welding, etc.
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SECTION 7: HOMEWORK & EXTENSION ACTIVITIES

Science Spy Challenge (All Students)

Find **ONE** chemical reaction at home tonight that involves heat. Record the following:

- Name of reaction: _____
- What gets hot or cold? _____
- Is it exothermic or endothermic? _____
- Signal word that helped you decide: _____

Share with the class the next day! Try to find something no one else will think of.

Extension Activities (Advanced / Gifted Students)

Activity	Description
Research Project	Find out how hand warmers work. Write 3-4 sentences explaining whether they are exothermic or endothermic and why.

Activity	Description
Creative Writing	Write a short story (10-12 sentences) from the point of view of a heat particle in an exothermic reaction.
Experiment Design	Design a safe at-home experiment to test whether mixing two kitchen ingredients produces heat or absorbs heat.
Chart Making	Draw a colourful chart showing 5 examples of each type of reaction with drawings and labels.

SECTION 8: STUDENT QUICK REFERENCE CARD

Print and distribute to students as a take-home reference card. Students can paste it in their science notebooks.

EXO = EXIT	ENDO = ENTER
Heat Exits → Gets HOT Signal words: hot, flame, burning, releases heat, gives off heat	Heat Enters → Gets COLD Signal words: cold, absorbs heat, cools down, needs heat
<u>Examples:</u> • Burning diya • Coal in sigri • Cooking dal • Rusting iron gate • Burning crackers	<u>Examples:</u> • Clay pot water cooling • Ice pack • Photosynthesis • Baking soda + vinegar • Cold packs